GATE-2004
BT: BIOTECHNOLOGY

Q. 1 – 10 carry one mark each

1. Expression of hundreds of different genes in DNA microarray technology is monitored by using
   (a) Radioactive probe
   (b) Visible chromogenic probe
   (c) UV absorbing probe
   (d) Fluorescent probe
2. Transfer of T - DNA from Ti plasmid into plant cell is mediated by
   (a) *mob* gene (b) *vir* gene
   (c) *nif* gene (d) *octopine* gene
3. For the growth of T-cell, the growth factor needed would be
   (a) Epidermal growth factor
   (b) Interleukin-2
   (c) Fibroblast growth factor
   (d) TNF- α
4. Nick translation of DNA is a method for making DNA probes. Identify from below what is NOT required for nick translation method
   (a) DNA polymerase
   (b) DNAase
   (c) Primers
   (d) Deoxyribonucleotides
5. During the functioning of biosensor which of the following sequences of events occurs
   (a) Enzymatic/cellular reaction → detector → transducer
   (b) Enzymatic/cellular reaction → transducer → detector
   (c) Enzymatic/cellular reaction → pressure gauge → time
   (d) Enzymatic/cellular reaction → vibrator → mechanical signal
6. During the media preparation for cultivation of cells, insoluble precipitates of calcium phosphates are often formed. Identify which method can be adopted to avoid this problem?
   (a) Hold the pH at 5.6
   (b) Hold the pH at 7.5
   (c) Add calcium salt first and then phosphate source
   (d) None the above
7. Somatic embryogenesis is a procedure in plant tissue culture methodology described best as
   (a) Formation of both shoot and root meristem
   (b) Formation of stable embryos
   (c) Formation of axillary buds
   (d) None of the above
8. An immobilized enzyme being used in a continuous plug flow reactor exhibits an effectiveness factor (η) of 1.2. The value of η being greater than one could be apparently due to one of the following reasons. Identify the correct reason.
   (a) The enzyme follows substrate inhibited kinetics with internal pore diffusion limitation
   (b) The enzyme experiences external film diffusion limitation
   (c) The enzyme follows sigmoidal kinetics
   (d) The immobilized enzyme is operationally unstable
9. The degree of inhibition for non-competitive inhibition of an enzyme catalyzed reaction
   (a) Increases with increases in substrate concentration
   (b) Reaches a maxima with increase in substrate concentration and then decreases
   (c) Is independent of substrate concentration
   (d) Decreases with increase in substrate concentration
10. The two columns given below indicate some of the fermentation products and the microbial cultures used for their production. Identify the correct set of groups from the four options.

Fermentation products	**Microbial cultures**
(A) Ethanol	(P) *Aspergillus niger*
(B) Streptomycin	(Q) *Zymomonas mobiles*
(C) Citric acid	(R) *Streptomyces griseus*
(D) Cellulase	(S) *Trichoderma reesei*

   (a) A-P; B-Q; C-R; D-S
   (b) A-Q; B-R; C-P; D-S
   (c) A-Q; B-R; C-S; D-P
   (d) A-S; B-R; C-Q; D-P

Q. 11 – 30 carry two marks each.

11. The culture fluids of 1000 to 5000 colonies of hybridoma are screened for monoclonal antibody by

P Western blot analysis

Q Antigen capture analysis

R Northern blot analysis

S Antibody capture analysis

Choose the correct pair from the following:

(a) P, Q (b) Q, R

(c) R, S (d) Q, S

12. Tobacco leaf discs are transfected with Agrobacterium tumefaciens strain containing binary vector (GUS as reporter gene) with selectable marker neo (kanamycin resistant gene) and then regenerated to plants. The plants are kanamycin resistant but leaf tissues are negative to GUS assay. The explanations are

(a) The plants are transformed for both genes by GUS gene is turned off

(b) The plants are transformed for only neo gene not the GUS gene

(c) The plants are not transformed at all, but the development of kanamycin resistance is due to somaclonal variation

(d) All of these

13. The restriction endonuclease HaeIII recognizes the sequence GG↓CC and the point of cleavage is given by the arrow. If you want to clone a piece of DNA in a plasmid digested by HaeIII, what will be restriction enzyme of choice ?

(a) SmaI (CC↓GGG) (b)NotI(GC↓GGCCGC)

(c) SalmIII (GG↓CC) (d)PstI (CTGCA↓G)

14. For the sequence of ds DNA given below, identify the set of primers required to amplify this DNA by PCR

3′ GACTCCA.....TACAACC 5′

CTGAGGT.....ATGTTGG 3′

(a) 5′ GGTTGTA and 5′ GACTCCA

(b) 5′ CTGAGGT and 5′ CCCAACAT

(c) 5′ ACTCAGT and 5′ ATGTTGG

(d) None of these

15. Expression of antisense RNA of ACC synthase in transgenic tomato plants inhibited the synthesis of ethylene resulting in

(a) Change in colour from green to red

(b) Change in aroma

(c) Change in colour from red to green

(d) None of these

16. Some of the genes from viruses introduced into plants in fully functional form often exhibit Mendelian inheritance because

(a) the genes are stably integrated in chromosomes

(b) the genes are stably maintained in vectors

(c) the genes are co-expressed with chromosomal genes

(d) the genes are not interrupted by introns

17. Agrobacterium based transformation of protoplasts obtained from dicots is based on the fact that

(a) These exhibit strong chromosomal structures

(b) These have two cotyledons

(c) These exhibit strong wound response

(d) These have long tap root system

18. What would be the effect of addition of 2,4-D on the production of berberine by cell culture of Thalictrum minus

(a) To stimulate growth and thereby urease secondary metabolite production

(b) Stimulate dedifferentiation and thereby decrease secondary metabolite production

(c) Stimulate proliferation and reduce secondary metabolite production

(d) None of these

19. Reverse vaccinology indicates

(a) From antigenic protein to vaccine development

(b) From antigenic polysaccharide to vaccine development

(c) From antibody to vaccine development

(d) From genome sequence to vaccine development

20. An enzyme following Michaelis – Menten kinetics with $V_m = 2.5$ mmol $m^{-3} s^{-1}$ and $K_m = 5.0$ mM was used to carry out the reaction in a batch stirred reactor. Starting with an initial substrate concentration of 0.1 M. The time required for 50% conversion of the substrate will be about

(a) 01 hr (b) 06 hr

(c) 02 hr (d) 12 hr

21. The maximum reaction velocity (V_m) for an enzyme catalyzed reaction was experimentally measured at two different temperatures and following results obtained :

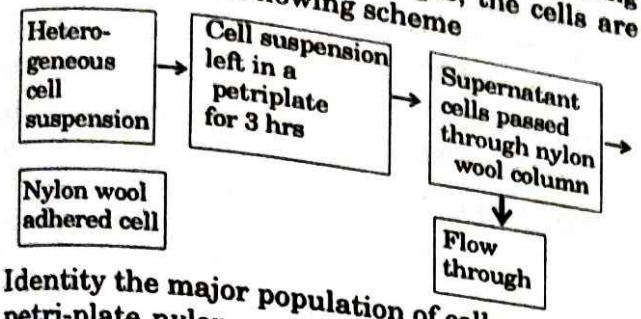
Temperature, °C	27	37
V_m mmol m^{-3}	2.25	4.50

The energy of activation for the reaction is

(a) 12834 cal mol^{-1} (b) 25668 cal mol^{-1}

(c) 6417 cal mol^{-1} (d) 19251 cal mol^{-1}

22. In a heterogeneous population of cells containing T-cells, B-cells and macrophages, the cells are separated in the following scheme



Identity the major population of cells present in petri-plate nylon wool adhered and nylon wool column flow through respectively

(a) Macrophage, B-cell, T-cell

(b) T-cell, B-cell, macrophage

(c) Macrophage, T-cell, B-cell

(d) B-cell, T-cell, macrophage

23. Match the following genetic elements with their functions.

Genetic elements	Functions
A. neo^R	1. Facilitates inducible expression of genes in eukaryotes
B. SV40	2. Facilitates constitutive expression of genes in eukaryotes
C. LTR	3. Allows amplification of gene
D. dhfr	4. Provides way of selecting eukaryotic cells, which have received foreign DNA

(a) A → 4, B → 2, C → 1, D → 3

(b) A → 2, B → 4, C → 1, D → 3

(c) A → 1, B → 4, C → 2, D → 3

(d) A → 4, B → 2, C → 3, D → 1

24. In the cell cycle of a typical eukaryote, the sequence of events operating at the time of cell division is

(a) S phase → G2 phase → G1 phase → M phase

(b) S phase → M phase → G1 phase → G2 phase

(c) S phase → G2 phase → M phase → G1 phase

(d) S phase → G1 phase → M phase → G2 phase

25. Batch fermentation of glucose to ethanol yields a productivity of 4.5 gl^{-1} hl^{-1}. If the yeast cell concentration in the fermentation brought is 5% (v/v) and the intracellular NAD/NADH concentration in the yeast cells in 10 μM the cycling rate of $NAD^+ \Leftrightarrow$ NADH will be

(a) 50,000 cycles hr^{-1} (b) 20,000 cycles hr^{-1}

(c) 100 cycles hr^{-1} (d) None of these

26. The kinetics of the disintegration of baker's yeast cells in a bead mill is described as $dp/dt = K.(P_m - P)$, where P is the concentration of protein released and P_m is the maximum protein concentration achievable. K is the first order rate constant and is 0.5 hr^{-1} for the system studied. The time required for the release of 90% of the intracellular proteins will be

(a) 10 hr (b) 2 hr

(c) 4.6 hr (d) None of these

27. Inversion of sucrose by immobilized invertase follows a substrate inhibited kinetics. The reaction rate (v) in mol m^{-3} hr^{-1} can be expressed as $V = 800\,[S]/\{400 + 50\,[S] + [S]^2\}$, where [S] is the sucrose concentration.

The immobilized invertase preparation is used in a CSTR with 100 mol m^{-3} sucrose concentration in the feed stream. If the reaction velocity passes through a maxima at [S] = 20 mol m^{-3}, the feed flow rate for a reactor volume of $1m^3$ to get the maximum productivity from the reactor should be

(a) 0.11 m^3 hr^{-1} (b) 1.10 $m^3 hr^{-1}$

(c) 5.05 m^3 hr^{-1} (d) None of these

28. Phytase an enzyme produced by Aspergillus niger can be adsorbed on microcrystalline cellulose powder (MCCP). The adsorption follows a Langmuir isotherm and the maximum concentration of the protein that can be obtained on the adsorbent is 70 mg cm^{-3}. At a concentration of 50mgl^{-1} of protein in the solution the concentration of protein on the adsorbent reaches 35 mg cm^{-3}. It is desired to recover 90% of the addition of MCCP to the solution. The concentration of the protein adsorbed on the solid at equilibrium will be

(a) 21.4 mg cm^{-3} (b) 214 mg cm^{-3}

(c) 2.14 mg cm^{-3} (d) None of these

29. Measurement of $K_l a$ in a bioreactor can be carried out by sodium sulfite oxidation method, that is based on the oxidation of sodium sulfite of sodium sulfate in the presence of a catalyst (Cu^{++} or Co^{++}). In a typical experiment, a laboratory fermenter was filled with 5 liter of 0.5 M sodium sulfite solution containing 0.003 MCu^{++} ions and the air was sparged in. After 10 minutes, the air flow was stopped and a 10 ml sample was taken and titrated. The concentration of sodium sulfite in the sample was found to be 0.20 M. The oxygen uptake rate for this aerated system will work out to be

(a) 0.08 gl^{-1} s^{-1} (b) 0.008 gl^{-1} s^{-1}

(c) 0.8 gl^{-1} s^{-1} (d) None of these

30. Examine the data given below in the table on purification of a protein X

Step	Volume (ml)	Protein concentration (mg/ml)	Enzyme activity (units/ml)
Crude cell-free extract	500	12.0	5.0
Ammonium sulfate precipitation	125	3.0	11.0
Ion-exchange Chromato-graphy	10	9.0	75.0

The yield percent and purification factor respectively at the end of the experiment will be approximately

(*a*) 25 and 3

(*b*) 30 and 20

(*c*) 20 and 30

(*d*) 75 and 30

ANSWERS

1. (*d*)	**2.** (*b*)	**3.** (*b*)	**4.** (c)	**5.** (*b*)	**6.** (*d*)	**7.** (*d*)	**8.** (*a*)	**9.** (*c*)	**10.** (*b*)
11. (*a*)	**12.** (*c*)	**13.** (*c*)	**14.** (*a*)	**15.** (*d*)	**16.** (*a*)	**17.** (*c*)	**18.** (*d*)	**19.** (d)	**20.** (*b*)
21. (*a*)	**22.** (*a*)	**23.** (*a*)	**24.** (*c*)	**25.** (*b*)	**26.** (*c*)	**27.** (*d*)	**28.** (*)	**29.** (*b*)	**30.** (*b*)

EXPLANATIONS

1. A DNA microarray (also commonly known as DNA chip or biochip) is a collection of microscopic DNA spots attached to a solid surface. Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome. Each DNA spot contains picomoles (10^{-12} moles) of a specific DNA sequence, known as *probes* (or *reporters* or *oligos*). These can be a short section of a gene or other DNA element that are used to hybridize a cDNA or cRNA (also called anti-sense RNA) sample (called *target*) under high-stringency conditions. Probe-target hybridization is usually detected and quantified by detection of fluorophore-, silver-, or chemiluminescence-labeled targets to determine relative abundance of nucleic acid sequences in the target.

2. Genes in the virulence region are grouped into the operons vir ABCDEFG, which code for the enzymes responsible for mediating transduction of T-DNA to plant cells

3. IL-2 is necessary for the growth, proliferation, and differentiation of T cells to become 'effector' T cells. IL-2 is normally produced by T cells during an immune response. Antigen binding to the T cell receptor (TCR) stimulates the secretion of IL-2, and the expression of IL-2 receptors IL-2R. The IL-2/IL-2R interaction then stimulates the growth, differentiation and survival of antigen-specific CD4+ T cells and CD8+ T cells As such, IL-2 is necessary for the development of T cell immunologic memory, which depends upon the expansion of the number and function of antigen-selected T cell clones.

4. This process is called nick translation because the DNA to be processed is treated with DNase to produce single-stranded "nicks." This is followed by replacement in nicked sites by DNA polymerase I, which elongates the 3' hydroxyl terminus, removing nucleotides by 5'-3' exonuclease activity, replacing them with dNTPs. Thus primers are not required.

5. the *sensitive biological element* (biological material (e.g. tissue, microorganisms, organelles, cell receptors, enzymes, antibodies, nucleic acids, etc.), a biologically derived material or biomimic component that interacts (binds or recognises) the analyte, the *transducer* or the *detector element* (works in a physicochemical way; optical, piezoelectric, electrochemical, etc.) that transforms the signal resulting from the interaction of the analyte with the biological element into another signal. Biosensor reader device with the associated electronics or signal processors that are primarily responsible for the display of the results in a user-friendly way.

6. The equilibrium solubility of four calcium salts (calcium oxalate hydrate, calcium citrate tetrahydrate, calcium phosphate, calcium glycerophosphate) were determined at controlled pH values (7.5, 6.0, 4.5 and < or = 3.0) and in distilled water.

7. Somatic embryogenesis is a process where a plant or embryo is derived from a single somatic cell or group of somatic cells. Somatic embryogenesis is thus not fully characterized by the given options. Hence none of the above is the right answer.

8. The effect of intraparticle diffusion in artificial membranes on the kinetic behavior of immobilized enzymes has been considered theoretically in terms of an effectiveness factor. When the immobilized enxymes obey a Michaelis-Menten relationship, an approximate equation is proposed for the effectiveness factor. For a substrate inhibited enzymatic reaction, the effectiveness factor may exceed unity and display multiple steady-state behavior. Thus, the overall rate of reaction in the artificial membrane is faster in this case than that when the substrate concentration in the interior is at the same as that at the exterior surface. In the case of a product inhibited enzymatic reaction, the effectiveness factor is always less than that which corresponds to a Michaelis-Menten relationship.

9. Non-competitive inhibitors are considered to be substances which when added to the enzyme alter the enzyme in a way that it cannot accept the substrate.

10. *A. niger* can ferment Citric Acid, Ethanol can be produced by Z. mobilis, Streptomycin is produced by S.griseus, Cellulose is produced by T. ressei.

11. Monoclonal antibodies bind to a single epitope within a target antigen. Western blot procedures allow for flexibility in choosing gel electrophoresis and blotting conditions, it is possible to modify buffers to retain enough higher order protein structure for detection by some antibodies. Thus can be brought to usage in screening for monoclonal antibodies.

12. The plants are not transformed at all, but the development of kanamycin resistance is due to the somaclonal variation.

13. The plasmid contains a genetic sequence corresponding to the recognition site of a restriction endonuclease, such as HaeIII. After foreign DNA fragments, which have also been cut with the same or similar restriction endonuclease, have been inserted into host cell, the restriction endonuclease gene is expressed by applying heat, or by introducing a biomolecule, such as arabinose.

14. The polymerase starts replication at the 3' end of the primer which means 5' end of the template and hence A is the correct option.

15. ACC synthase catalyzes the synthesis of ACC, a precursor for ethylene. Ethylene serves as a hormone in plants for stimulating or regulating the ripening of fruit, the opening of flowers, and the abscission (or shedding) of leaves and hence none of the given options is suitable.

16. Mendelian inheritance is observed as the viral genes have been stably integrated and thus expresses itself.

17. Agrobacterium infection method has been extensively used to transfer foreign DNA into a number of dicotlyledonous species with the exception of soybean (Glycine max). However, this technique of gene transfer was not successful in monocotyledons for reasons yet unknown most probable being the lack of wound response of monocotyledonous cells.

18. Adding of 2,4-D (1 M), in the cultured cells grew rapidly, producing little berberine. On the other hand, the berberine-producing activity was remarkably enhanced by simultaneous administration of auxin and cytokinin, although cell growth was inferior.

19. Reverse vaccinology is development of vaccine that employs bioinformatics. It indicates, from genome sequence to vaccine development and hence the vaccine is better and more specific.

20. By MM equation $V = V_m.S/K_m + S$.

21. $$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1\,T_2}\right)$$

$$\log(2) = \frac{E_a(10)}{2.303 \times 1.986(300 \times 310)}$$

$$E_a = \frac{0.3010}{10} \times 425359.5$$

$$= 12803.3 \text{ cal mol}^{-1}$$

22. B-cells selectively adhere to the fiber of nylon wool while T-cells do not and hence found in column flow resulting in successful recovery of both cell types. Whereas macrophage attaches to the petri plates and hence its major population will be there.

23. neo^R stands for the eukaryotic gene expression marker, SV40 facilitate gene expression in eukaryotes, LTR induces new gene expression and dhfr is a gene amplification factor.

24. In eukaryotes, the cell cycle consists of four discrete phases: G_1, S, G_2, and M. The **S** or **synthesis phase** is when DNA replication occurs, and the **M** or **mitosis phase** is when the cell actually divides. The other two phases — G_1 and G_2, the so-called **gap phases** — are

less dramatic but equally important. During G_1, the cell conducts a series of checks before entering the S phase. Later, during G_2, the cell similarly checks its readiness to proceed to mitosis.

25. P = 4.5 gm/l.hr Ethanol production yeast cell conc.(X) = 5% v/v

$NAD^+/NADH$ ceme in yeast cell = 10 μm (m = ml/lit)

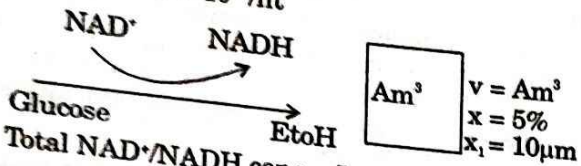
= (me/ωt) the cycling rate of $NAD^+ \rightleftharpoons NADH$ will be.

$X = \frac{5}{100}$ ml in 1 liter of broth,

$NAD^+/NADH = 10^{-5}$ ml/lit.

$\frac{50}{100}\ \frac{ml}{lit} = 10^{-5}$ ml/lit $\frac{\text{No. of cydest}}{\text{unit time}}$ = cycle rate.

5×10^{-5}/lit



Total $NAD^+/NADH$ conc = 5×10^{-5}(v/v)

Now, cycle rate = $\left(\frac{1}{5\times10^{-5}}\right)\frac{\text{cycle}}{hr}$

= 20,000 cycle/hr.

26. $\frac{dP}{dt} = K(P_m - P)$

$$K = \frac{2.303}{t}\log_{10}\left(\frac{100P}{10P}\right)$$

$$K = \frac{2.303}{t}$$

$$t = \frac{2.303}{0.5} \Rightarrow 4.606 \text{ hour}$$

27. $V = \frac{800[s]}{400+50[s]+[s]^2}$

$S_0 = 100$ mol/m³ → $V_0 = 5.195$ ml/m³ hr.

i.e. V_{max} when S = 20 ml/m³,

$\therefore V_{max} = 8.889$ ml/m³.hr. $V = \left(\frac{dp}{dt}\right) D = \frac{F}{V}$

Now, $K_m = \frac{V_{max}}{2} = 4.444$

Now,

$V_0 = V_{max}\frac{S}{K_m + S} = 7.27$ ml/m³. hr